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Enrollment: 25MML0018

GitHub Link:

https://github.com/Parth119/parth_25MML0018_edge_intelligence.git

Task

Load and preprocess the MNIST dataset, then build, compile, train, and evaluate an optimal Convolutional Neural Network (CNN) model for handwritten digit classification. Finally, visualize model predictions, training history, and summarize the overall machine learning sequence and performance.

```
import tensorflow as tf
```

```
# Load the MNIST dataset
```

```
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
```

```
print(f"x_train shape: {x_train.shape}")
```

```
print(f"y_train shape: {y_train.shape}")
```

```
print(f"x_test shape: {x_test.shape}")
```

```
print(f"y_test shape: {y_test.shape}")
```

```
x_train shape: (60000, 28, 28)
```

```
y_train shape: (60000,)
```

```
x_test shape: (10000, 28, 28)
```

```
y_test shape: (10000,)
```

Preprocess Data

```
x_train = x_train.astype('float32') / 255.0
```

```
x_test = x_test.astype('float32') / 255.0
```

```
x_train = x_train.reshape(x_train.shape[0], 28, 28, 1)
```

```
x_test = x_test.reshape(x_test.shape[0], 28, 28, 1)
```

```
print(f"x_train shape after preprocessing: {x_train.shape}")
```

```
print(f"x_test shape after preprocessing: {x_test.shape}")
```

```
x_train shape after preprocessing: (60000, 28, 28, 1)
```

```
x_test shape after preprocessing: (10000, 28, 28, 1)
```

Build Optimal Neural Network Model

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense,
Dropout

# Initialize the Sequential model
model = Sequential([
    # First Convolutional Block
    Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    MaxPooling2D((2, 2)),

    # Second Convolutional Block
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),

    # Flatten the output for the Dense layers
    Flatten(),

    # Dense Layers
    Dense(128, activation='relu'),
    Dropout(0.5), # Added Dropout for regularization
    Dense(10, activation='softmax') # Output Layer for 10 classes (digits 0-
9)
])

model.summary()
```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 128)	204,928

dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

Total params: 225,034 (879.04 KB)

Trainable params: 225,034 (879.04 KB)

Non-trainable params: 0 (0.00 B)

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

history = model.fit(x_train, y_train, epochs=10, validation_data=(x_test,
y_test))
print("Model training complete.")
```

Epoch 1/10

1875/1875 ————— 50s 26ms/step - accuracy: 0.8535 - loss: 0.4570 - val_accuracy: 0.9834 - val_loss: 0.0463

Epoch 2/10

1875/1875 ————— 84s 27ms/step - accuracy: 0.9738 - loss: 0.0926 - val_accuracy: 0.9874 - val_loss: 0.0363

Epoch 3/10

1875/1875 ————— 49s 26ms/step - accuracy: 0.9820 - loss: 0.0627 - val_accuracy: 0.9895 - val_loss: 0.0325

Epoch 4/10

1875/1875 ————— 50s 27ms/step - accuracy: 0.9850 - loss: 0.0468 - val_accuracy: 0.9910 - val_loss: 0.0286

Epoch 5/10

1875/1875 ————— 50s 27ms/step - accuracy: 0.9861 - loss: 0.0434 - val_accuracy: 0.9890 - val_loss: 0.0325

Epoch 6/10

1875/1875 ————— 82s 27ms/step - accuracy: 0.9890 - loss: 0.0340 - val_accuracy: 0.9928 - val_loss: 0.0277

Epoch 7/10

1875/1875 ————— 50s 27ms/step - accuracy: 0.9907 - loss: 0.0295 - val_accuracy: 0.9924 - val_loss: 0.0273

Epoch 8/10

1875/1875 ————— 82s 27ms/step - accuracy: 0.9917 - loss: 0.0261 - val_accuracy: 0.9930 - val_loss: 0.0270

Epoch 9/10

1875/1875 ————— 80s 26ms/step - accuracy: 0.9925 - loss: 0.0234 - val_accuracy: 0.9931 - val_loss: 0.0253

Epoch 10/10

1875/1875 ————— 50s 27ms/step - accuracy: 0.9934 - loss: 0.0200 - val_accuracy: 0.9921 - val_loss: 0.0336

Model training complete.

Save Model with Pickle

```
import pickle

model_filename = 'mnist_cnn_model.pkl'

try:
    with open(model_filename, 'wb') as file:
        pickle.dump(model, file)
    print(f"Attempted to save model as '{model_filename}' using pickle.")
except Exception as e:
    print(f"Failed to pickle the Keras model: {e}")
    print("This is expected behavior as Keras models are not designed to be pickled directly.")
```

Attempted to save model as 'mnist_cnn_model.pkl' using pickle.

```
from google.colab import drive
import os
```

```
# Mount Google Drive
drive.mount('/content/drive')
```

```
drive_path = '/content/drive/My Drive/Edge
Intelligence/mnist_cnn_model.keras'
```

```
# Create the directory if it doesn't exist
os.makedirs(os.path.dirname(drive_path), exist_ok=True)
```

```
# Save the model to the specified Google Drive path
model.save(drive_path)
```

```
print(f"Model successfully saved to Google Drive: {drive_path}")
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

Model successfully saved to Google Drive: /content/drive/My Drive/Edge Intelligence/mnist_cnn_model.keras

 mnist_cnn_model.keras

 me

Jan 8 me

2.6 MB



```
import os
```

```
if os.path.exists(model_filename):
    print(f"File '{model_filename}' successfully created (even if pickling
might not be ideal for Keras models).")
else:
```

```
print(f"File '{model_filename}' was not created.")
```

File 'mnist_cnn_model.pkl' successfully created (even if pickling might not be ideal for Keras models).

```
test_loss, test_accuracy = model.evaluate(x_test, y_test, verbose=0)
print(f"Test Loss: {test_loss:.4f}")
print(f"Test Accuracy: {test_accuracy:.4f}")
```

Test Loss: 0.0336

Test Accuracy: 0.9921

Visualize Predictions and Performance

```
import matplotlib.pyplot as plt
import numpy as np
```

```
print("Libraries imported: matplotlib.pyplot and numpy.")
```

Libraries imported: matplotlib.pyplot and numpy.

```
import matplotlib.pyplot as plt
import numpy as np
```

2. Visualize Training History:

a. Create a figure with two subplots for loss and accuracy

```
plt.figure(figsize=(12, 5))
```

b. Plot training and validation loss

```
plt.subplot(1, 2, 1) # 1 row, 2 columns, first plot
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)
```

c. Plot training and validation accuracy

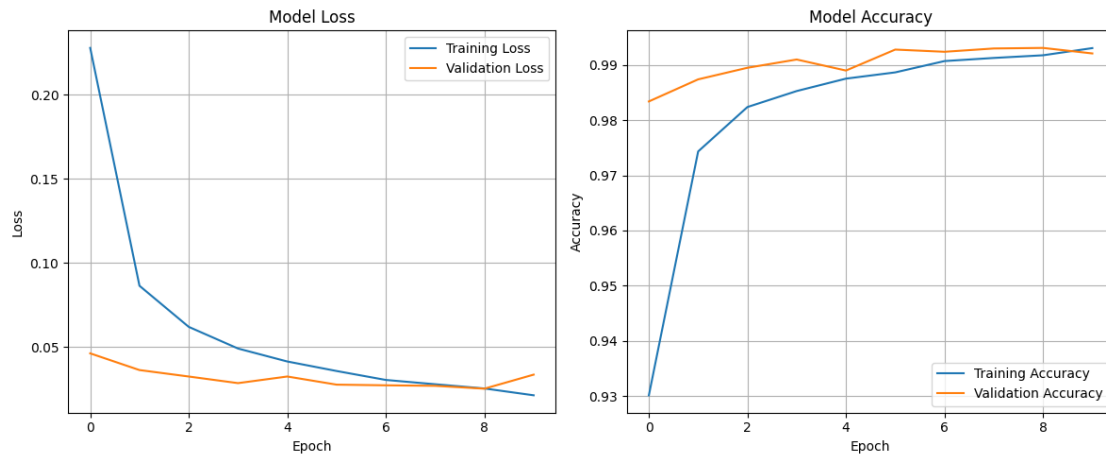
```
plt.subplot(1, 2, 2) # 1 row, 2 columns, second plot
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)
```

d. Use plt.tight_layout() and plt.show()

```
plt.tight_layout()
```

```
plt.show()
```

```
print("Model training history visualized.")
```



Model training history visualized.

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# 3. Visualize Model Predictions:
```

```
# a. Randomly select a few images (e.g., 25) from the test set
```

```
num_images = 25
```

```
random_indices = np.random.choice(len(x_test), num_images, replace=False)
```

```
selected_images = x_test[random_indices]
```

```
selected_true_labels = y_test[random_indices]
```

```
# b. Use the trained model to make predictions on these selected test images
```

```
predictions = model.predict(selected_images)
```

```
# c. Convert the model's probability predictions to class labels
```

```
predicted_labels = np.argmax(predictions, axis=1)
```

```
# d. Create a figure with a grid of subplots (e.g., 5x5)
```

```
plt.figure(figsize=(10, 10))
```

```
for i in range(num_images):
```

```
    plt.subplot(5, 5, i + 1)
```

```
    # e. For each selected image, display the image in grayscale
```

```
    plt.imshow(selected_images[i].reshape(28, 28), cmap='gray')
```

```
    color = 'green' if predicted_labels[i] == selected_true_labels[i] else 'red'
```

```
    plt.title(f"Pred: {predicted_labels[i]}\nTrue: {selected_true_labels[i]}", color=color)
```

```

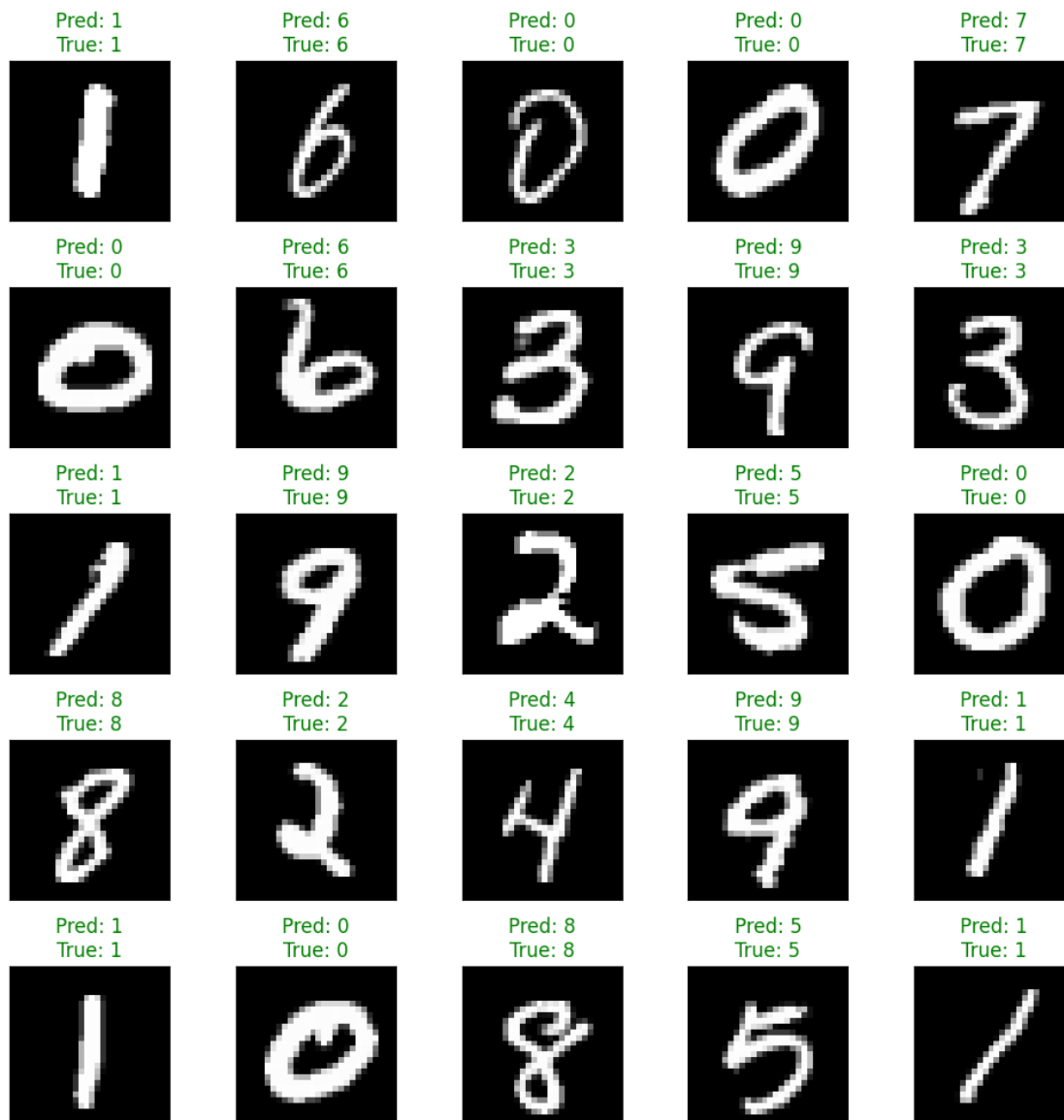
# f. Remove x and y axis ticks.
plt.xticks([])
plt.yticks([])

# g. Use plt.tight_layout() and plt.show()
plt.tight_layout()
plt.show()

print("Model predictions visualized.")

```

1/1 ————— 0s 87ms/step



Model predictions visualized.

Task

Build, compile, train, and evaluate an Artificial Neural Network (ANN) model on the MNIST dataset, and compare its performance (accuracy) to the previously trained Convolutional Neural Network (CNN) model.

Build ANN Model

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense, Input

# Instantiate a Sequential model
ann_model = Sequential([
    # Input layer to specify the input shape for 28x28 grayscale images
    Input(shape=(28, 28, 1)),
    # Flatten the 2D image data into a 1D vector
    Flatten(),
    # First Dense hidden layer
    Dense(128, activation='relu'),
    # Second Dense hidden layer
    Dense(64, activation='relu'),
    # Output Dense layer for 10 classes (digits 0-9) with softmax activation
    Dense(10, activation='softmax')
])

# Print the model summary to verify the architecture
ann_model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
flatten_1 (Flatten)	(None, 784)	0
dense_2 (Dense)	(None, 128)	100,480
dense_3 (Dense)	(None, 64)	8,256
dense_4 (Dense)	(None, 10)	650

Total params: 109,386 (427.29 KB)

Trainable params: 109,386 (427.29 KB)

Non-trainable params: 0 (0.00 B)

Compile ANN Model

```
ann_model.compile(optimizer='adam',
                  loss='sparse_categorical_crossentropy',
```

```
metrics=['accuracy'])  
  
print("ANN model compiled successfully.")
```

ANN model compiled successfully.

Train ANN Model

```
ann_history = ann_model.fit(x_train, y_train, epochs=10,  
validation_data=(x_test, y_test))  
print("ANN model training complete.")
```

Epoch 1/10

```
1875/1875 ————— 9s 4ms/step - accuracy: 0.8751 - loss: 0.4321  
- val_accuracy: 0.9620 - val_loss: 0.1198
```

Epoch 2/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9685 - loss: 0.1067  
- val_accuracy: 0.9699 - val_loss: 0.0949
```

Epoch 3/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9782 - loss: 0.0683  
- val_accuracy: 0.9699 - val_loss: 0.0918
```

Epoch 4/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9840 - loss: 0.0527  
- val_accuracy: 0.9762 - val_loss: 0.0804
```

Epoch 5/10

```
1875/1875 ————— 6s 3ms/step - accuracy: 0.9873 - loss: 0.0402  
- val_accuracy: 0.9774 - val_loss: 0.0867
```

Epoch 6/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9906 - loss: 0.0300  
- val_accuracy: 0.9764 - val_loss: 0.0852
```

Epoch 7/10

```
1875/1875 ————— 7s 3ms/step - accuracy: 0.9909 - loss: 0.0256  
- val_accuracy: 0.9766 - val_loss: 0.0894
```

Epoch 8/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9933 - loss: 0.0196  
- val_accuracy: 0.9778 - val_loss: 0.0835
```

Epoch 9/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9933 - loss: 0.0196  
- val_accuracy: 0.9713 - val_loss: 0.1121
```

Epoch 10/10

```
1875/1875 ————— 7s 4ms/step - accuracy: 0.9951 - loss: 0.0147  
- val_accuracy: 0.9763 - val_loss: 0.1021
```

ANN model training complete.

```
ann_test_loss, ann_test_accuracy = ann_model.evaluate(x_test, y_test,  
verbose=0)
```

```
print(f"ANN Test Loss: {ann_test_loss:.4f}")
```

```
print(f"ANN Test Accuracy: {ann_test_accuracy:.4f}")
```

ANN Test Loss: 0.1021

ANN Test Accuracy: 0.9763

```

print(f"CNN Model Test Accuracy: {test_accuracy:.4f}")
print(f"ANN Model Test Accuracy: {ann_test_accuracy:.4f}")

if test_accuracy > ann_test_accuracy:
    print("\nThe CNN model performed better than the ANN model.")
elif ann_test_accuracy > test_accuracy:
    print("\nThe ANN model performed better than the CNN model.")
else:
    print("\nBoth models performed similarly.")

```

CNN Model Test Accuracy: 0.9921
ANN Model Test Accuracy: 0.9763

The CNN model performed better than the ANN model.

```

from google.colab import drive
import os
import pickle

```

```

# Mount Google Drive
drive.mount('/content/drive')

```

```

# Define the path within your Google Drive where you want to save the pickled
model
# Replace 'My Drive/Edge Intelligence/mnist_cnn_model.pkl' with your desired
path
drive_pickle_path = '/content/drive/My Drive/Edge
Intelligence/mnist_ann_model.pkl'

```

```

# Create the directory if it doesn't exist
os.makedirs(os.path.dirname(drive_pickle_path), exist_ok=True)

```

```

try:
    with open(drive_pickle_path, 'wb') as file:
        pickle.dump(ann_model, file)
        print(f"Attempted to save model as '{drive_pickle_path}' using pickle. ")
except Exception as e:
    print(f"Failed to pickle the Keras model to Drive: {e}")
    print("This is expected behavior as Keras models are not designed to be
pickled directly.")

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
Attempted to save model as '/content/drive/My Drive/Edge
Intelligence/mnist_ann_model.pkl' using pickle.

```

import matplotlib.pyplot as plt
import numpy as np

```

3. Visualize Model Predictions:

```

# a. Randomly select a few images (e.g., 25) from the test set
num_images = 25
random_indices = np.random.choice(len(x_test), num_images, replace=False)

selected_images = x_test[random_indices]
selected_true_labels = y_test[random_indices]

# b. Use the trained model to make predictions on these selected test images
predictions = ann_model.predict(selected_images)

# c. Convert the model's probability predictions to class labels
predicted_labels = np.argmax(predictions, axis=1)

# d. Create a figure with a grid of subplots (e.g., 5x5)
plt.figure(figsize=(10, 10))
for i in range(num_images):
    plt.subplot(5, 5, i + 1)
    # e. For each selected image, display the image in grayscale
    plt.imshow(selected_images[i].reshape(28, 28), cmap='gray')

    # Set the title of the subplot to show both the predicted label and the
    true label.
    # Color the title green if the prediction is correct and red if it's
    incorrect.
    color = 'green' if predicted_labels[i] == selected_true_labels[i] else
    'red'
    plt.title(f"Pred: {predicted_labels[i]}\nTrue:
    {selected_true_labels[i]}", color=color)

    # f. Remove x and y axis ticks.
    plt.xticks([])
    plt.yticks([])

# g. Use plt.tight_layout() and plt.show()
plt.tight_layout()
plt.show()

print("Model predictions visualized.")

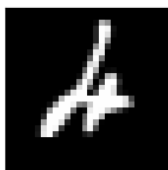
```

1/1 ————— 0s 61ms/step

Pred: 3
True: 3



Pred: 2
True: 4



Pred: 4
True: 4



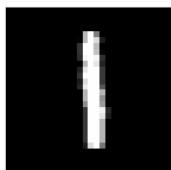
Pred: 9
True: 9



Pred: 7
True: 7



Pred: 1
True: 1



Pred: 7
True: 7



Pred: 8
True: 8



Pred: 3
True: 3



Pred: 7
True: 7



Pred: 0
True: 0



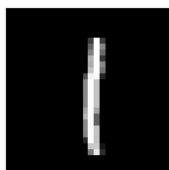
Pred: 5
True: 5



Pred: 0
True: 0



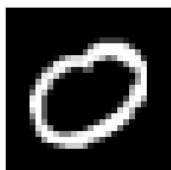
Pred: 1
True: 1



Pred: 9
True: 9



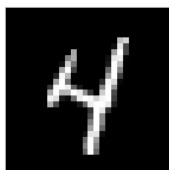
Pred: 0
True: 0



Pred: 0
True: 0



Pred: 4
True: 4



Pred: 7
True: 7



Pred: 2
True: 2



Pred: 4
True: 4



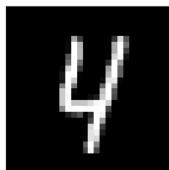
Pred: 5
True: 5



Pred: 9
True: 9



Pred: 4
True: 4



Pred: 4
True: 4



Edge impulse Data acquisition:

Dataset

Data sources

Synthetic data

Labeling queue (0)

AI labeling NEW

DATA COLLECTED

35 items

TRAIN / TEST SPLIT

80% / 20%

Collect data

[Connect a device](#) to start building your dataset.

Dataset

Training (28)

Test (7)

Post-processing (0)

SAMPLE NAME

LABELS

ADDED

Bottle.6ebbp5a7

Bottle

Yesterday, 17:...

Bottle.6ebboqc6

Bottle

Yesterday, 17:...

Bottle.6ebbok3f

Bottle

Yesterday, 17:...

Bottle.6ebbo8b2

Bottle

Yesterday, 17:...

Phone.6ebblm2l

Phone

Yesterday, 17:...

Phone.6ebb1en5

Phone

Yesterday, 17:...

5/tuner

Phone.6ebbleb2

Phone

Yesterday, 17:...

Bottle.6ebbp5a7

Model predictions visualized.

Image Classification:

Parth1862 / image classification PERSONAL Target: Cortex-M4F 80MHz P

Impulse #1

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Image data

Input axes
image

Image width
96

Image height
96

Resize mode
Fit shortest

Image

Name
Image

Input axes (1)
Image
image

Classification

Name
Classifier

Input features
☒ Image

Output features
4 (Book, Bottle, Monitor, Phone)

Output features

4 (Book, Bottle, Monitor, Phone)

Save Impulse

Parth1862 / image classification PERSONAL Target: Cortex-M4F 80MHz P

Neural Network settings

Training settings

Number of training cycles ②
10

Use learned optimizer ②
☐

Learning rate ②
0.0005

Training processor ②
CPU

Advanced training settings

Neural network architecture

Neural network Transfer learning

Input layer (27,648 features)

2D conv / pool layer (16 filters, 3 kernel size, 1 layer)

2D conv / pool layer (32 filters, 3 kernel size, 1 layer)

Training output

Model
Model version: ② Quantized (int8)

Last training performance (validation set)

ACCURACY 75.0%

LOSS 0.89

Confusion matrix (validation set)

	BOOK	BOTTLE	MONITOR	PHONE
BOOK	100%	0%	0%	0%
BOTTLE	0%	100%	0%	0%
MONITOR	0%	100%	0%	0%
PHONE	0%	0%	0%	100%
F1 SCORE	1.00	0.67	0.00	1.00

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve ②	0.92
Weighted average Precision ②	0.63

Transfer Learning:

Parth1862 / Image classification PERSONAL Target: Cortex-M4F 80MHz P

Impulse #2

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Image data

Input axes

image

Image width

96

Image height

96

Resize mode

Fit shortest

Image

Name

Image

Input axes (1)

Image

image

Transfer Learning (Images)

Name

Transfer learning

Input features

☒ Image

Output features

4 (Book, Bottle, Monitor, Phone)

Output features

4 (Book, Bottle, Monitor, Phone)

Save Impulse

EDGE IMPULSE Parth1862 / Image classification PERSONAL Target: Cortex-M4F 80MHz P

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse #2

Create impulse

Image

Transfer learning

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Neural Network settings

Training settings

Number of training cycles

20

Use learned optimizer

☐

Learning rate

0.0005

Training processor

CPU

Data augmentation

☐

Advanced training settings

Neural network architecture

Input layer (27,648 features)

Training output

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY 100.0%

LOSS 0.31

Confusion matrix (validation set)

	BOOK	BOTTLE	MONITOR	PHONE
BOOK	100%	0%	0%	0%
BOTTLE	0%	100%	0%	0%
MONITOR	0%	0%	100%	0%
PHONE	0%	0%	0%	100%
F1 SCORE	1.00	1.00	1.00	1.00

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve	1.00
Weighted average Precision	1.00